

'OOps'

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Course: Object Oriented Programming t

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Note:

I'm submitting it late because my moellim was blocked

<https://github.com/ISHAQKHAN196/OOP-2nd-Semester-Lab-Task>

TASK:01

A BANK WANTS A SYSTEM THAT:

- STORES CUSTOMER INFORMATION
- MANAGES BANK ACCOUNT OPERATIONS
- ALLOWS DEPOSIT, WITHDRAWAL, AND BALANCE CHECKING

WE WILL USE TWO CLASSES:

1. CUSTOMER
2. BANK ACCOUNT

```
public class Main1 {  
    public static void main(String[] args) {  
        Customer c1 = new Customer();  
        c1.name = "ISHAQ-KHAN";  
        c1.customerId = 73423; // ISHAQKHAN-73423  
        BankAccount acc1 = new BankAccount();  
        acc1.accountNumber = 76585;  
        acc1.balance = 999999999;  
        acc1.customer = c1; // ISHAQKHAN-73423  
        acc1.displayAccount();  
        acc1.deposit();  
        acc1.withdraw();  
        acc1.displayAccount();  
    }  
}
```

MainClass:

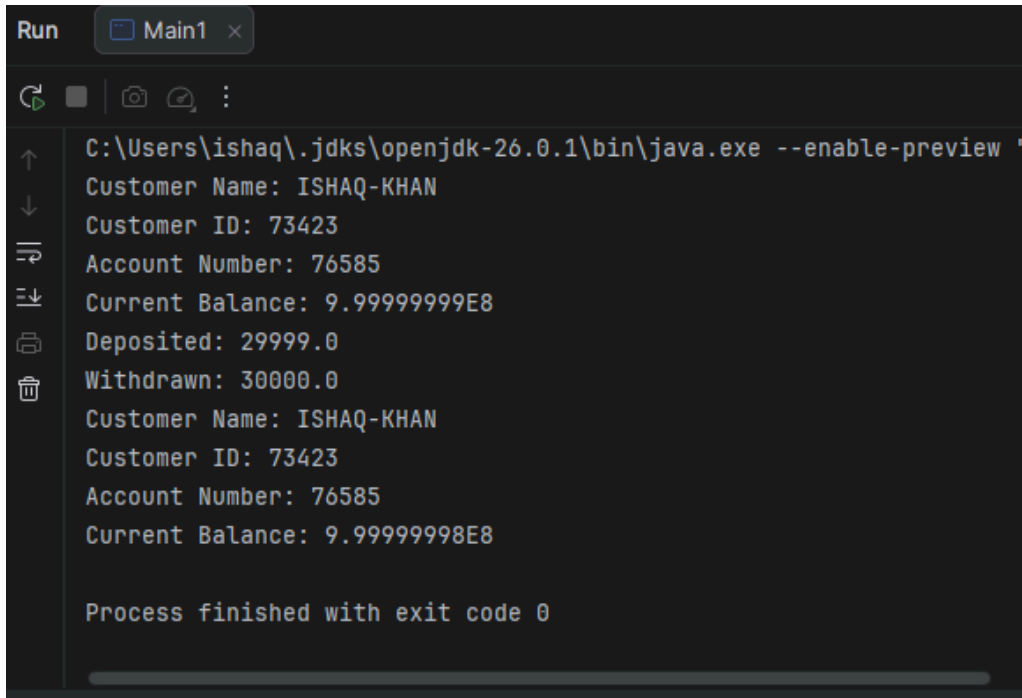
```
public class Customer { // ISHAQKHAN-73423
    String name;
    int customerId;
    void displayCustomer() {
        System.out.println("Customer Name: " + name);
        System.out.println("Customer ID: " + customerId);
    } // ISHAQKHAN-73423
}
```

Class customer:

```
public class BankAccount {
    int accountNumber;
    double balance; // ISHAQKHAN-73423
    Customer customer; // ISHAQKHAN-73423
    void deposit() {
        balance = balance + (double) 29999;
        System.out.println("Deposited: " + (double) 29999); //
        ISHAQKHAN-73423
    }
    void withdraw() {
        if ((double) 30000 <= balance) { // ISHAQKHAN-73423
            balance = balance - (double) 30000;
            System.out.println("Withdrawn: " + (double)
            30000); // ISHAQKHAN-73423
        } else {
            System.out.println("Insufficient Balance!");
        }
    }
    void displayAccount() {
        customer.displayCustomer();
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Current Balance: " + balance);
    }
}
```

Class BankAccount

output:



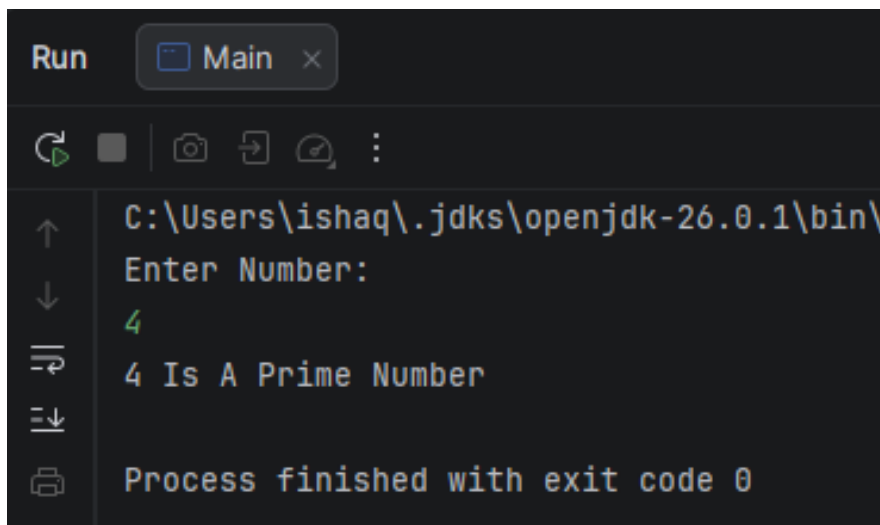
The screenshot shows a 'Run' console window with a tab labeled 'Main1'. The console displays the following output:

```
C:\Users\ishaq\.jdk\openjdk-26.0.1\bin\java.exe --enable-preview "  
Customer Name: ISHAQ-KHAN  
Customer ID: 73423  
Account Number: 76585  
Current Balance: 9.99999999E8  
Deposited: 29999.0  
Withdrawn: 30000.0  
Customer Name: ISHAQ-KHAN  
Customer ID: 73423  
Account Number: 76585  
Current Balance: 9.99999998E8  
  
Process finished with exit code 0
```

TASK:02;

Prime numbers in java

```
void main() {  
    Scanner s = new Scanner(System.in);  
    IO.println("Enter Number:");  
    int n;  
    int count = 0;  
    n = s.nextInt();  
    for (int i = 1; i < n; i++) {  
        if (n % i == 0)  
            count++;  
    }  
    if (count == 2)  
        IO.println(n + " Is A Prime Number");  
    else  
        IO.println(n + " Is Not A Prime Number");  
}
```

Code:**output:**

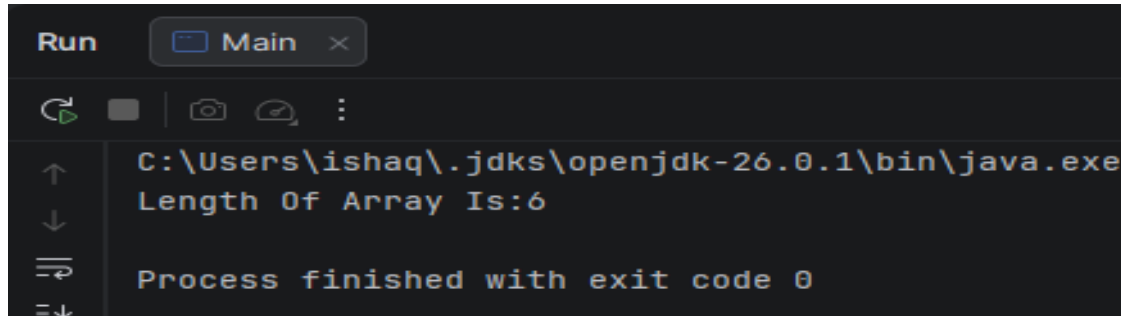
The screenshot shows a 'Run' window titled 'Main' with a dark background. It contains the following text:

```
C:\Users\ishaq\.jdk\openjdk-26.0.1\bin\j  
Enter Number:  
4  
4 Is A Prime Number  
  
Process finished with exit code 0
```

TASK:03;

Find length of Array;

```
public class Main {  
  
    public static void main(String[] args){  
        int[] a ={1,2,8,6,9,10};  
        int length;  
        length=a.length;  
        System.out.println("Length Of Array Is:"+length);  
    }  
}
```

A screenshot of a Java IDE's 'Run' window. The window title is 'Run' with a sub-tab 'Main'. It shows the command 'C:\Users\ishaq\.jdk\openjdk-26.0.1\bin\java.exe' and the output 'Length Of Array Is:6'. Below the output, it states 'Process finished with exit code 0'. The window includes standard IDE icons for running, debugging, and other actions.

```
Run Main x  
C:\Users\ishaq\.jdk\openjdk-26.0.1\bin\java.exe  
Length Of Array Is:6  
Process finished with exit code 0
```

Output

TASK:04;

Find the number is palindrome or not

```
import java.util.Scanner;
// ISHAQKHAN-73423// ISHAQKHAN-73423// ISHAQKHAN-73423
public class Main {
    public static void main(String[] args){// ISHAQKHAN-73423
        Scanner s=new Scanner(System.in);
        int num,temp,r=0;
        System.out.println("Enter Number:");
        num=s.nextInt();
        temp=num;// ISHAQKHAN-73423
        while(num>0){
            r=r*10+(num%10);
            num/=10;
        }// ISHAQKHAN-73423
        if(temp==r)
            System.out.println(temp+" Is Palindrome");// ISHAQKHAN-73423
        else
            System.out.println(temp+" Is Not Palindrome");
    }
}
```

OUTPUT;

```
C:\Users\ishaq\.jdk\openjdk-26.0.1\bin\java.
Enter Number:
7876
7876 Is Not Palindrome

Process finished with exit code 0
```

TASK:05;

Find the factorial number through Function;

```
import java.util.Scanner;
// ISHAQKHAN-73423
public class Main { // ISHAQKHAN-73423
    static void fact() {
        Scanner s = new Scanner(System.in); // ISHAQKHAN-73423
        int a;
        System.out.println("Enter Number For Factorial:");
        a = s.nextInt(); // ISHAQKHAN-73423
        int fact = 1;
        for (int i = 1; i <= a; i++) // ISHAQKHAN-73423
            fact *= i; // ISHAQKHAN-73423
        System.out.println("Factorial Of "+a+" Is = "+fact);
    }
    public static void main(String[] args) {
        fact();
    } // ISHAQKHAN-73423
}
```

Output

```
Enter Number For Factorial:
454
Factorial Of 454 Is = 0

Process finished with exit code 0
|
```

TASK:06;

Swapping numbers using tow variables

```
import java.util.Scanner;
// ISHAQKHAN-73423
public class Main {
    public static void main(String[] args){// ISHAQKHAN-73423
        Scanner s=new Scanner(System.in);// ISHAQKHAN-73423
        int a,b;// ISHAQKHAN-73423
        System.out.println("Enter Values For Swap:");
        a=s.nextInt();
        b=s.nextInt();// ISHAQKHAN-73423
        a=a+b;
        b=a-b;
        a=a-b;// ISHAQKHAN-73423
        System.out.println("Values After Swap Are:");
        System.out.println("a= "+a);// ISHAQKHAN-73423
        System.out.println("b= "+b);
    }
}
```

Output;

```
C:\Users\ishaq\.jdk\openjdk-26.0.1\bin\java.exe
Enter Values For Swap:
566
00
Values After Swap Are:
a= 0
b= 566

Process finished with exit code 0
```


TASK:07;

Swap values using third variable

```
import java.util.Scanner;// ISHAQKHAN-73423
public class Main {
    public static void main(String[] args){
        Scanner s=new Scanner(System.in);
        int a , b, temp;// ISHAQKHAN-73423
        System.out.println("Enter Values For Swap:");
        a=s.nextInt();// ISHAQKHAN-73423
        b=s.nextInt();
        temp=a;// ISHAQKHAN-73423
        a=b;// ISHAQKHAN-73423
        b=temp;// ISHAQKHAN-73423
        System.out.println("Values After Swap Are:");
        System.out.println("a= "+a);
        System.out.println("b= "+b);
    }
}
```

Output;

```
C:\Users\ishaq\.jdk\openjdk-26.0.1\bin\java.exe "-jav
Enter Values For Swap:
34
3
Values After Swap Are:
a= 3
b= 34

Process finished with exit code 0
```

TASK:08

Right angle triangle using nested loop

```
public class Main {  
    // ISHAQKHAN-73423  
    public static void main(String[] args) {  
        for (int i = 1; i <= 5; i++) { // ISHAQKHAN-73423  
            for (int j = 1; j <= i; j++) { // ISHAQKHAN-73423  
                System.out.print("*");  
            }  
            System.out.println(); // ISHAQKHAN-73423  
        }  
    }  
    // ISHAQKHAN-73423  
}
```

Output;

```
C:\Users\ishaq\.jdk\openjdk-26.0.1\bin\java.exe --enabl  
*  
**  
***  
****  
*****  
  
Process finished with exit code 0
```